

PRISM: A Pedagogical Resource Inspection System for Multimodal Educational Content Quality Evaluation

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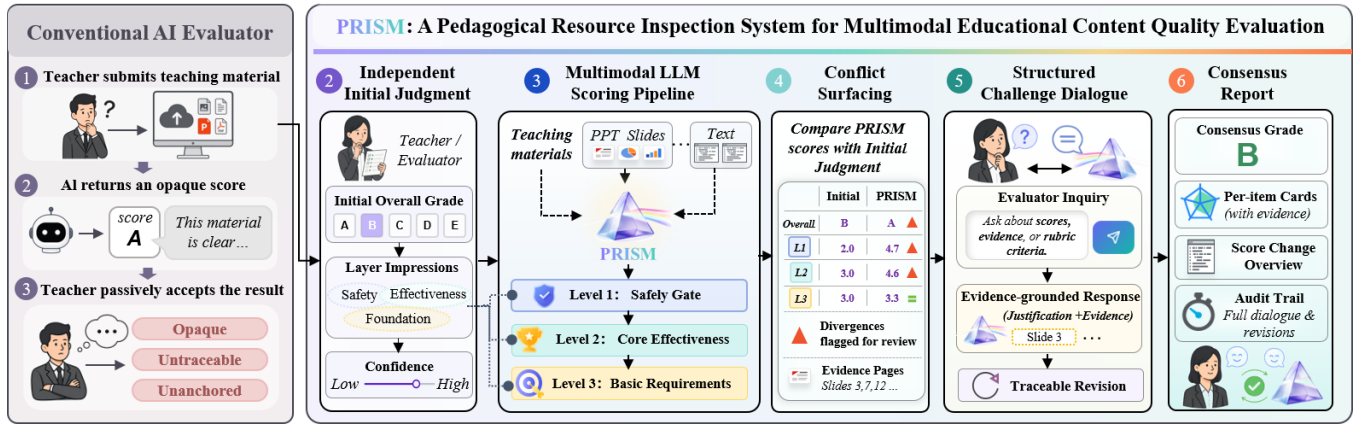


Figure 1: System overview. A conventional AI evaluator (left) returns an opaque score the teacher passively accepts; PRISM (right) scaffolds a five-stage workflow: independent initial judgment, multimodal LLM scoring against a three-layer non-compensatory rubric, conflict surfacing, structured challenge dialogue, and an audited consensus report.

Abstract

We present PRISM (Pedagogical Resource Inspection System for Multimodal content), an interactive system for evaluating AI-generated and teacher-created multimodal educational resources. PRISM grounds evaluation in a validated 24-indicator three-layer non-compensatory rubric: a safety gate with veto authority, a core effectiveness layer yielding a five-tier grade, and a diagnostic layer. A multimodal LLM

jointly processes slide images and text with page-level evidence. Beyond opaque scoring, PRISM supports traceable human-AI challenge: evaluators record an independent initial judgment before scores are revealed and may contest any score through dialogue. Project page: <https://chenzhangze-web.github.io/prism-page/>.

CCS Concepts

• Computing methodologies → Artificial intelligence; • Applied computing → Education.

Keywords

multimodal LLM; educational resource evaluation; human-in-the-loop; non-compensatory rubric

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1 Introduction

Generative AI has compressed lesson-plan creation from hours to minutes, but the quality of AI-generated materials varies widely in accuracy, pedagogical soundness, and ethical compliance [9]. Applying a general-purpose LLM as evaluator produces holistic judgments that conflate safety-critical failures (fabricated citations, undisclosed AI authorship) with mild pedagogical weaknesses; rubric-guided LLM scoring improves alignment with human judges but does not enforce safety asymmetry [3, 4]. Neither approach addresses the multimodal nature of modern teaching materials, where image-text consistency and media quality require joint visual-linguistic reasoning that recent multimodal LLM research has begun to address for documents [5, 11] and unified image-text understanding [6].

PRISM contributes (1) a theory-grounded 24-indicator rubric that enforces a non-compensatory veto so safety-critical failures cannot be offset by pedagogical strengths; (2) a multimodal LLM scoring engine that reasons jointly over slide images and text with page-level evidence; and (3) a five-dimensional challenge mechanism, building on proactive conversational AI [2], in which an *independent initial judgment* seeds a structured-challenge dialogue, operationalising appropriate-reliance principles for educational-resource evaluation (Figure 1).

2 Evaluation Framework

PRISM organises evaluation along a 24-indicator three-layer non-compensatory rubric (Figure 1), grounded in Merrill’s First Principles [8], Mayer’s CTML [7], the ICAP framework [1], and UNESCO AI-in-education guidelines [9], and validated through a Modified Delphi study with 15 domain experts. **L1 Safety Gate** (6 indicators covering factual accuracy, ethics, privacy, link safety, citation verifiability, and AI-authorship disclosure) screens for content correctness and ethical compliance: any L1 score ≤ 2 triggers an immediate “prohibited from use” verdict regardless of L2/L3 performance, operationalising the fatal-flaw principle [10]. **L2 Core Effectiveness** (11 indicators on teaching aims, content coverage, learning activities, cognitive design, and assessment) is the grade-determining layer, with the indicator mean mapped to grades A/B/C/D/E (thresholds 4.5 / 3.5 / 2.5 / 1.5). **L3 Basic Requirements** (7 indicators on visual design, readability, media quality, metacognitive support, and technical usability) contributes diagnostic detail without affecting the grade. This non-compensatory layering is the key architectural decision distinguishing PRISM from holistic LLM scoring, which has no equivalent veto.

3 System Design

Architecture and Workflow. PRISM is implemented as a Next.js front end with a Python/FastAPI back end. The five-step workflow proceeds as: (i) resource upload with metadata, used to conditionally activate indicators; (ii) *independent initial judgment* (overall A to E grade, three-layer impressions, and a confidence rating) recorded *before* any AI output is shown; (iii) three sequential LLM rounds, one per layer, with real-time progress; (iv) a comparison

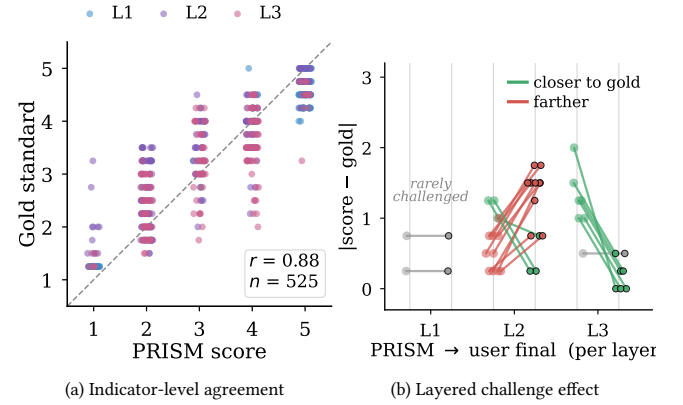


Figure 2: Pilot evaluation by rubric layer. (a) PRISM vs. 4 experts gold-standard scores. (b) Challenge-induced shifts from PRISM to user-final scores.

view that surfaces conflicts and supports the challenge phase (Section 3); (v) a final report combining overall-grade cards, a three-column qualitative summary (strengths, weaknesses, suggestions), and per-indicator detail panels with embedded challenge records.

Multimodal LLM Scoring. Each evaluation round receives *both* slide images and extracted text, since indicators on multimodal consistency, visual quality, and readability cannot be judged from text alone. Three sequential LLM calls (one per layer) each receive layer-specific rubric prompts with 1 to 5 behavioral anchors per indicator and return structured JSON (code, score, comment, evidence_pages). Separating layers keeps each layer independently auditable.

Five-Dimensional Challenge Mechanism. After AI scoring, PRISM enters a challenge phase organised along five dimensions (Figure 1): (1) **Conflict Surfacing** highlights divergences ≥ 1 grade level or ≥ 1.5 points; (2) **Challenge Modality** offers structured and free-form contestation; (3) **Response Strategy** requires each revision to be grounded in concrete evidence, such as a specific slide page and rubric anchor; (4) **Authority Locus** preserves both AI and user-final scores for audit; (5) **Learning Loop** retains challenge history, surfacing recurring revision targets.

4 Preliminary Evaluation

We conducted a pilot study with $N=12$ educators (3 in-service teachers, 9 graduate students), each evaluating two stage-matched resources from a curated 12-resource subset covering 5 quality tiers (A to E) and 8 subject domains. The gold standard was set by two parallel 4-expert panels with combined ICC[2,k] = 0.919. PRISM matched the gold standard exactly on 62.5% of sessions and within ± 1 grade tier on all of them; per-indicator MAE was 0.643 (Pearson $r=0.88$, $p < .001$). Validity varies systematically across layers (L1: 0.380; L2: 0.676; L3: 0.869). The 22 challenge-induced revisions diverge sharply across layers (Figure 2): L3 revisions are uniformly corrective (7/7 closer to gold), L2 revisions skew toward user over-strictness (9/12 farther), and L1 is essentially unchallenged. SUS averaged 82.92 (Good on Bangor’s scale) and TAM subscales 5.96 to 6.08 on a 7-point scale.

Acknowledgments

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